

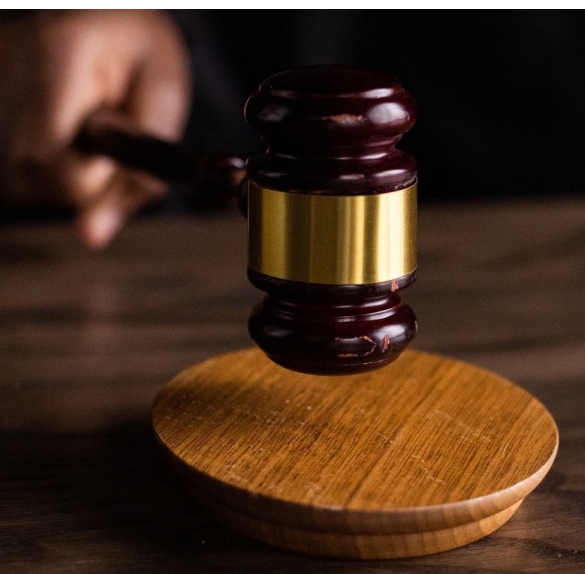
Foundations of Cyber Security

COMSM 0118 - Legal and Ethical Aspects

What is law?

“... a binding custom or practice of a community: a rule of conduct or action prescribed or formally recognised as binding or enforced by a controlling authority.”

“... a system of rules that a society or government develops in order to deal with crime, business agreements, and social relationships.”



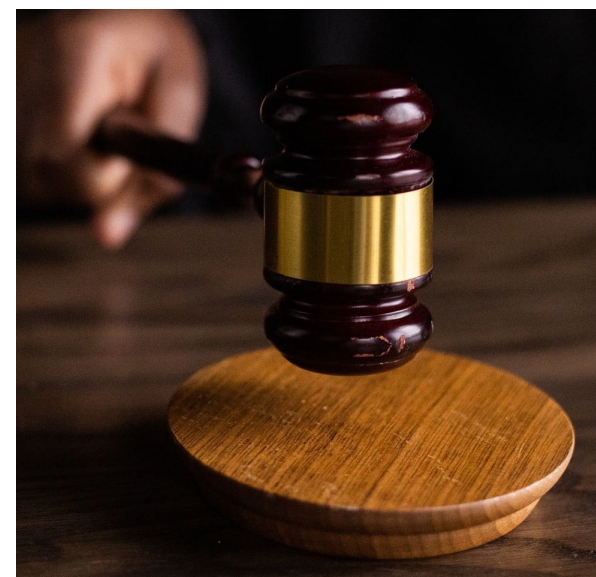
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Purpose of the law...

- Addressing the responsibility of persons and the disposition of property.
- Protect people/society
- Regulates the human conduct



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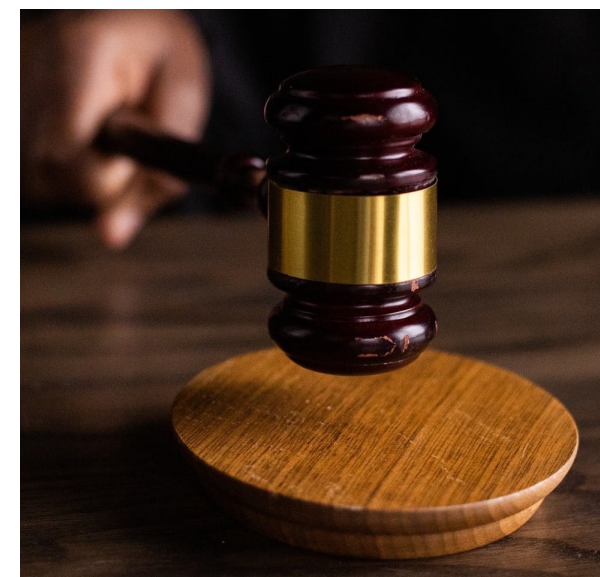
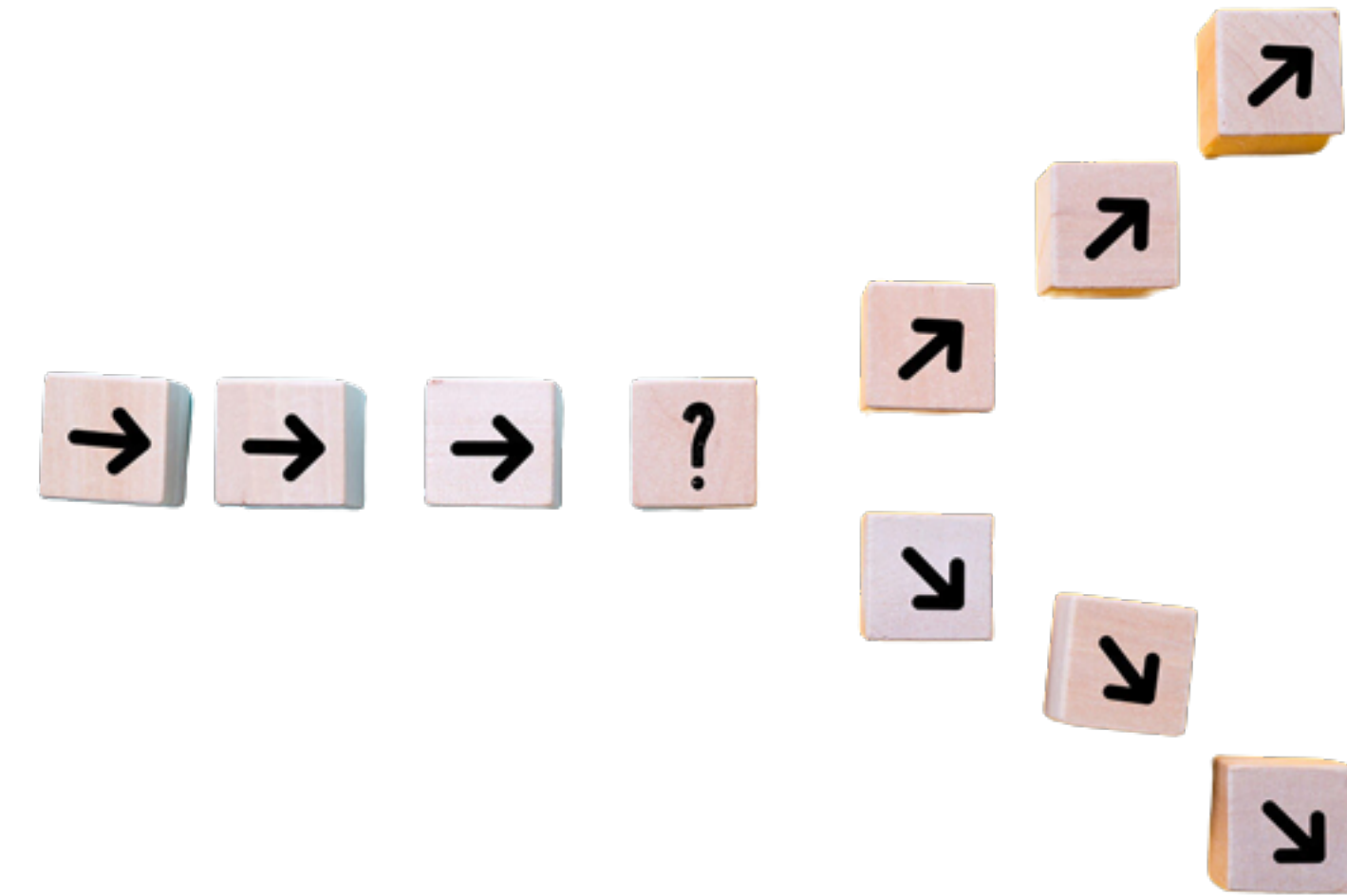
Sources of the law...

- Custom
- Actions of representatives of people
- Decisions of the court (Precedent)
- Actions of administrative agency
- Agreements between people



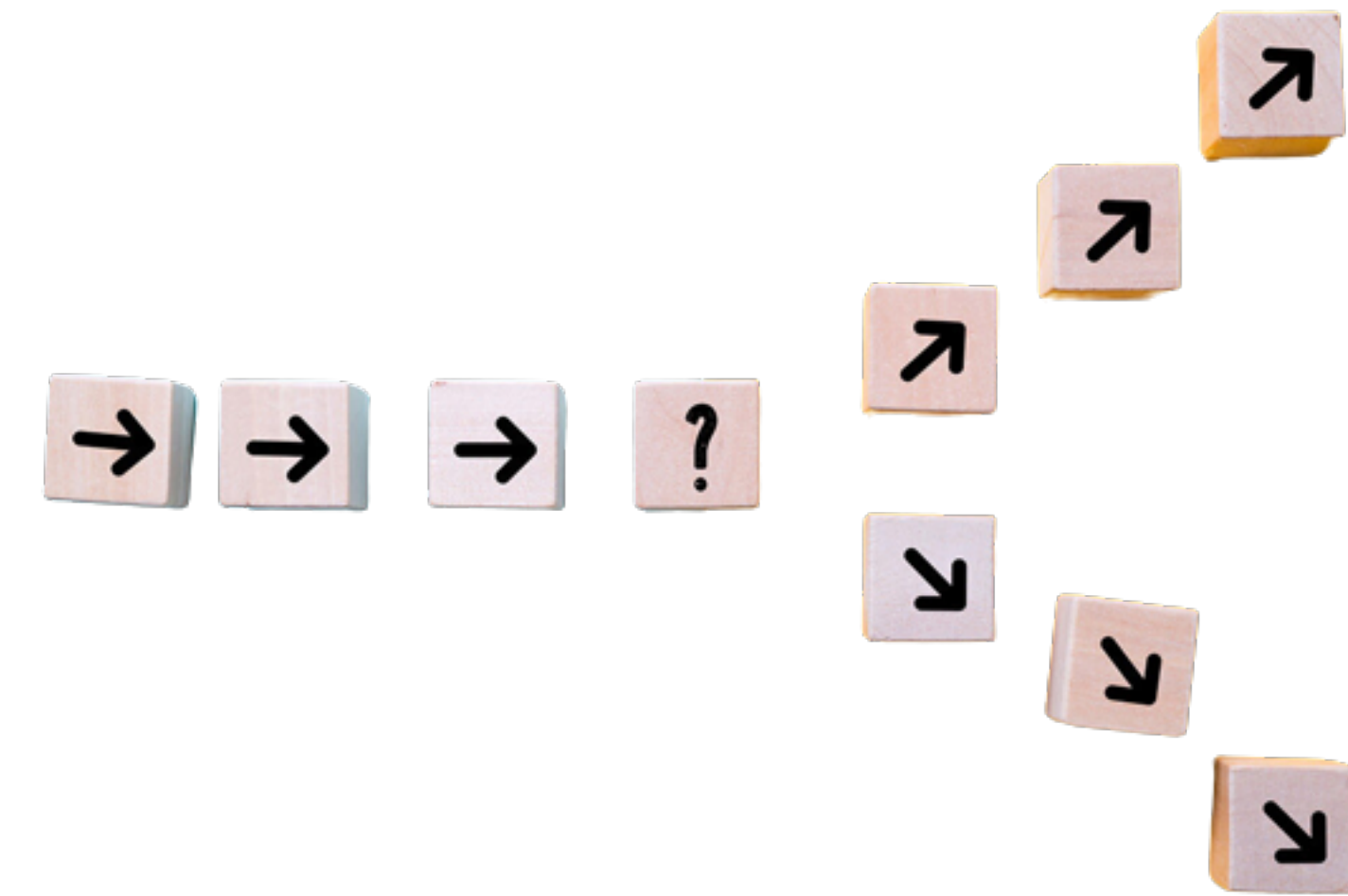
What is law?

- *is dynamic and influences society*
- *changes all the times (e.g., ruling)*
- *People influence the law*
- *New ideas (technology)*
- *can interact in with other laws in complex ways*
- *Law is territorial; a reflection of the society.*



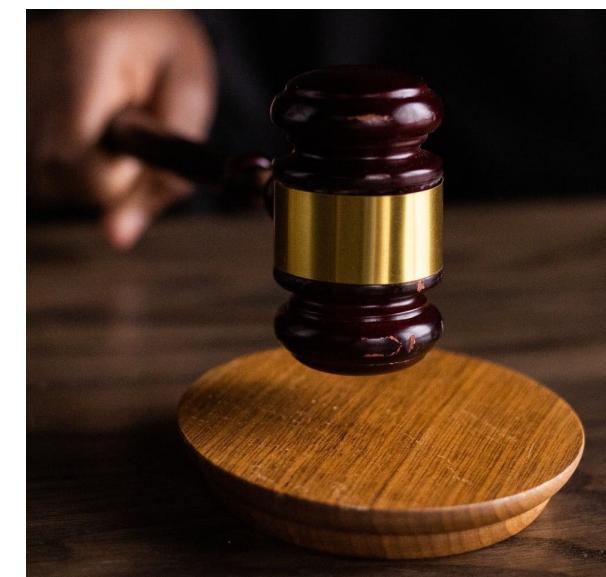
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Uncertainty...

- *Method of interpretation are different **
- *Differing sources **



* Robert Carolina - CyBOK: Law and Regulations

Law and technology

Cyberspace is unbounded, is not a territory.

The internet get us to discuss international laws.

Because sometimes we need to resolve issues of two entities arguing between two countries.

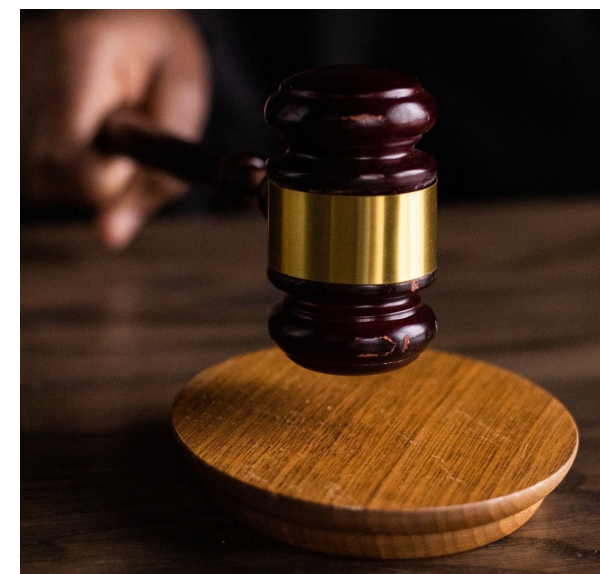
In case of two people arguing, each country will apply their own laws to protect their own people.



Law and technology

Examples...

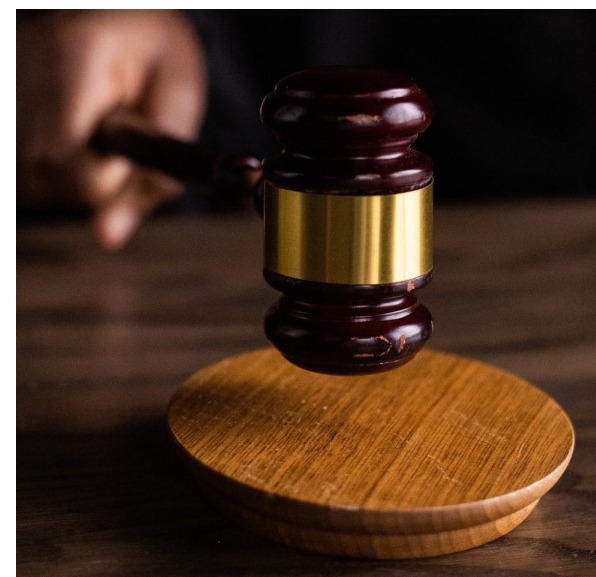
- General Data Protection Regulation (GDPR)
 - Consumer Privacy Protection Act (CPPA)
 - Children's Online Privacy Protection Act (COPRA)
 - California Consumer Privacy Act (CCPA)
 - Computer Fraud And Abuse Act (CFAA)
 - Computer Misuse Act (CMA)
 - Health insurance Portability and Accountability Act (HIPPA)
 - Online Safety bill
-
- Intellectual Property (IP) laws
 - Copyright laws
 - Contract laws
 - Tort laws



Law and Security Practitioners

Potential risky activities...

- Vulnerability research
- Penetration testing
- Reverse engineering
- Sensitive information
- ...



Law and Security Practitioners

Potential risky activities...

- Vulnerability research
 - Penetration testing
 - Reverse engineering
 - Sensitive information
 - ...
-
- Accessing devices without owner's permission
 - Copying, modifying, or running software that belongs to other people (copyright laws)
 - Circumventing measures designed to prevent or restrict access to software
 - Transferring non-published information, code, or equipment related to cryptography to a foreign destination, outside the ordinary course of research.
 - disclosing information about software or a system design that a company keeps secret from a competitor



Law: Computer Misuse Act (CMA)

History...

“The Appeal Court in 1985 indicated that if a person sent a message from London to divert funds from New York to his accounts in Geneva, the theft would not have taken place in London and so English courts would not have had jurisdiction to try the offender.” [1]

Enacted in 1990

*“After two individuals obtained **without knowledge** the username and password of an engineer who worked for a telecom company. They managed to get **access** to Prince Philip’s emails. They were arrested, convicted under the Forgery and Counterfeiting Act (1981). Later appealed because they stated that they didn’t do it for financial gain but to demonstrate that the system could be **hacked**. Convictions were overturned.”*

Computer Misuse Act 1990 “created to deal with those who use or access computers for malicious purposes” or hacking

[1] <https://www.lawteacher.net/free-law-essays/criminal-law/computer-misuse-act.php?vref=1>

Law: Computer Misuse Act (CMA)

Focuses on the following sections...

- Unauthorised access to computer material.
- Unauthorised access with intent to commit or facilitate commission of further offences.
- Unauthorised acts with intent to impair, or with recklessness as to impairing, operation of computer.
- Unauthorised acts causing, or creating risk of, serious damage.
- Making, supplying or obtaining articles for use in offence under section 1, 3 or 3ZA.

*“It criminalises several acts, including **accessing data without authorisation** (i.e., hacking), **installing malware** (e.g., computer viruses, spyware, or ransomware) on a person's computer, and **producing materials** for committing mentioned offences.”*

Law: Computer Misuse Act (CMA)

What is a “Computer?”

“ Any device or group of interconnected or related devices, one or more of which pursuant to a program, performs automatic processing of data” *Council of Europe Cybercrime Convention 2001*

“... a device for storing, processing and retrieving information.” *Court Cases: DPP v McKeown, DPP v Jones*

Desktop computers, laptops, IoT devices, mobile devices, and servers.



Law: Computer Misuse Act (CMA)

Jurisdiction (territorial scope)

... requires proof of at least one ‘**significant link**’ with the ‘**home country**’

Home country: ***England, Wales, Scotland and North Ireland ****

Section 1, 3, 3ZA

A significant link could include:



- The accused is in the home country at the time of the offence
- The target of the CMA offence is in the home country
- The technological activity which has facilitated the offending may have passed through a server based in the home country

Section 5, in relation to an offence under Section 3ZA, significant link

- That the accused was in the home country concerned at the time when s/he committed the unauthorised act (or caused it to be done);
- That the unauthorised act was done in relation to a computer in the home country concerned;
- That the unauthorised act caused, or created a significant risk of, serious damage of a material kind (within the meaning of that section) in the home country concerned.

<https://www.cps.gov.uk/legal-guidance/computer-misuse-act>

* <https://www.legislation.gov.uk/ukpga/1990/18/crossheading/jurisdiction>

Law: Computer Misuse Act (CMA)

Examples...

DPP v Bignell [1998]

Two police officers (Mr and Mrs Bignell), who were authorised to request information from the police national computer (PNC) for policing purposes only, requested a police computer operator to obtain information from the PNC which, unbeknown to the operator, was for their own personal use. [1]

R v Vallor [2003]

A 22-old Welsh Web designer who pleaded guilty to creating and distributing three mass mailer viruses was sentenced to two years imprisonment at Southwark Crown Court this afternoon. [2]

R v Mudd [2018] 1 Cr App R (S) 33 (7)

The offender, who was aged between 16 and 18 over the course of the offending, admitted offences under sections 1 and 3, and a further offence of concealing criminal property. He had devised a distributed denial of service program which he used on some occasions himself and on other occasions supplied the program for payment for others to use. In total, 1.7 million DDoS attacks were carried out directed at well over half a million individual IP addresses or domain names. [3]

[1] DPP v Bignell [1998] 1 Cr. App. R. 1

[2] https://www.theregister.com/2003/01/21/welsh_virus_writer_vallor_jailed/

[3] <https://www.cps.gov.uk/legal-guidance/computer-misuse-act>

Law: Computer Misuse Act (CMA)

Critics...

- Was enacted before cyber security existed
- It criminalises security researchers
- It fails to recognise the value of "ethical" or "white hat" hacking
- It hard to prove “intent”
- No scope for hacking-back
- Web-Scraping deemed lawful

Law: Computer Misuse Act (CMA)

Critics...



Security researcher [@robbykedotcom](#) recently discovered and responsibly disclosed security vulnerabilities to [@AppertaUK](#) about sensitive information stored on their publicly accessible repositories. He now faces legal retaliation. Let's help him.



gofundme.com
Defend Rob's Responsible Disclosure, organized by Rob Dyke
TL/DR: I made a responsible security disclosure to Apperta, an NHS tech org, that had ... Rob Dyke needs your support for Defend Rob's Responsible ...

Dyke found a public repo containing the source code for an insecure online portal and its database containing usernames, hashed passwords, email addresses, and API keys. [1]

Laws and regulations

Other issues...

- Restrictions on exporting security technologies
- Strong cryptographic products (e.g., Export Admin Regulations)
- Matters/Issues classified as secret by a state

Further Reading

- **CyBOK Law and regulations: https://www.cybok.org/media/downloads/Law_Regulation_v1.0.2.pdf**
- **J. Goldsmith and T. Wu, Who Controls the Internet? Illusions of a Borderless World. New York, NY: Oxford University Press, 2006.**
- **S. K Hayes and J. Niehaus, Defensive tactics for Today's Law enforcement, Taylor & Francis Group, 2017**
- **Computer Misuse Act: <https://www.cps.gov.uk/legal-guidance/computer-misuse-act>**